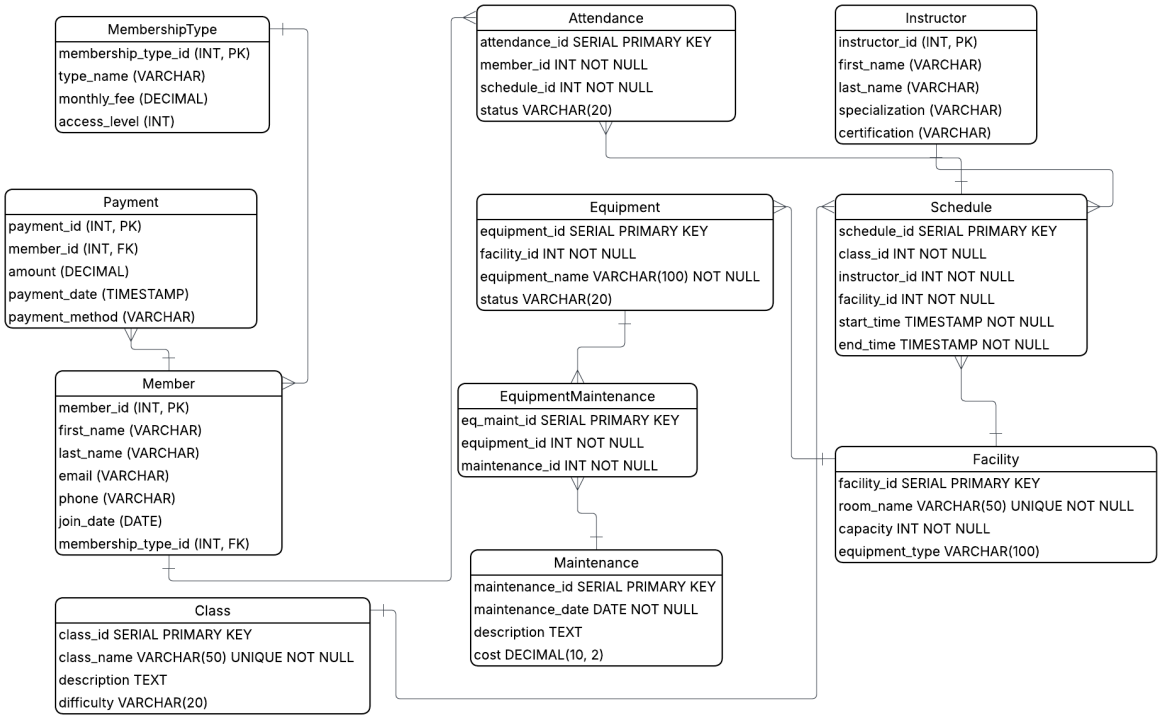
1.1 Logical Scheme



1.2 Objects

Table Description(members)

The central repository for all registered clients, storing their personal details and linking them to a specific membership plan.

Table Name	Field name	Field Description	Data Type / Constraints

members	member_id	Unique identifier for each gym member.	Int, PK, Auto increment
	first_name	The member's given name.	VARCHAR(50), not null
	last_name	The member's family name.	VARCHAR(50), not null
	date_of_birth	The member's birth date.	DATE
	membership_type_id	Reference to the specific plan the member has purchased.	INT, Not null, FK

Comments on table relationships

- Many-to-One with membership_types (each member follows one plan).
- One-to-Many with payments (one member can have multiple transaction records).
- One-to-Many with schedule_enrollment (one member can sign up for multiple sessions).

Example with data

member_id	first_name	last_name	date_of_birth	membership_type_id
101	John	Doe	1990-05-15	1
102	Jane	Smith	1985-11-20	2
103	Robert	Brown	1998-02-03	1

Table Description

A lookup table (dictionary) that stores the available subscription plans offered by the gym, including their costs, durations, and specific access permissions.

Table Name	Field name	Field Description	Data Type / Constraints
membership_types	membership_type_id	Unique identifier for each membership plan.	Int, PK, Auto increment
	name	The name of the membership tier (e.g., "Basic", "Premium").	VARCHAR(50), not null
	duration_months	The validity period of the membership in months.	int, not null
	price	The cost of the membership plan.	Decimal(10,2), not null
	access_level	Defines which areas or services the member can use.	VARCHAR(50)

Comments on table relationships

Comments on table relationships: This is a parent table to **members**. It has a One-to-Many relationship with the **members** table, as one membership type can be assigned to many different members.

Example with data

membership_type_id	name	duration_months	price	access_level
1	Gold Membership	12	599.99	Full Access

2	Standard Monthly	1	49.99	Gym Only
3	Student Pass	3	99.00	Gym & Pool

Table description (Classes)

A catalog of the workout programs and services offered (e.g., Yoga, HIIT), defining what each class entails without specific timing.

Table Name	Field name	Field Description	Data Type / Constraints
classes	class_id	Unique identifier for a type of workout program.	Int, PK, Auto increment
classes	name	The title of the class (e.g., "Morning Yoga").	VARCHAR(100), not null
classes	description	A summary of what the class involves.	TEXT

Comments on table relationships: Has a One-to-Many relationship with the **schedule** table. One class definition (e.g., "Zumba") can appear multiple times in the weekly schedule.

Examples

class_id	name	description
20	Vinyasa Flow	A dynamic yoga class focusing on breath and movement.
21	Power Cardio	High-intensity interval training for fat burning.
22	Core Strength	Focuses on abdominal and back muscle stability.

Table description **(Facilities)**

An inventory of the gym's physical spaces (studios, pools, gyms) used to manage maximum occupancy and room locations.

Table Name	Field name	Field Description	Data Type / Constraints
facilities	facility_id	Unique identifier for a specific room or area.	Int, PK, Auto increment
	name	The name of the room (e.g., "Studio A", "Pool").	VARCHAR(50), not null
	capacity	Maximum number of people allowed in the area at once.	Int, not null
	location_description	Details on where the facility is located within the building.	VARCHAR(100)

Comments on table relationships: Has a One-to-Many relationship with the **schedule** table. A single facility can host many different scheduled classes throughout the day.

facility_id	name	capacity	location_description
5	Studio A	25	Second floor, near the elevators
6	Main Gym	50	Ground floor, west wing
7	Aquatic Center	20	Basement level

Table Description (Payments)
A financial log that records every transaction made by members, used to track revenue and individual payment history.

Table Name	Field name	Field Description	Data Type / Constraints
payments	payment_id	Unique identifier for each transaction.	Int, PK, Auto increment

	amount	Total money paid in the transaction.	DECIMAL(10,2), NOT NULL
	payment_date	Timestamp of when the payment was processed.	DATETIME
	payment_method	The mode of payment (e.g., "Credit Card", "Cash").	VARCHAR(30)
	member_id	Reference to the member who made the payment.	INT, not null, FK

Comments on table relationships: Has a Many-to-One relationship with **members**. Every payment must be linked to a specific member ID to track who paid the amount.

payment_id	amount	payment_date	payment_method	member_id
5001	599.99	2024-01-10 14:30:00	Credit Card	101
5002	49.99	2024-01-11 9:15:00	Cash	102

Table Description (Schedule)

The operational heart of the system that schedules a specific class in a specific room with a specific instructor at a set time

Table Name	Field name	Field Description	Data Type / Constraints
schedule	schedule_id	Unique identifier for a specific time slot in the calendar.	Int, PK, Auto increment
	class_id	Reference to the program being taught.	INT, not null, FK
	instructor_id	Reference to the trainer leading this session.	INT, not null, FK
	facility_id	Reference to the room where the session takes place.	INT, not null, FK

	duration	The specific time or length of the session.	VARCHAR(30), not null

Comments on table relationships: Many-to-One with **classes**, **instructors**, and **facilities**. One-to-Many with **schedule_enrollment** (one scheduled slot can have many members signed up for it).

schedule_id	class_id	instructor_id	facility_id	duration
301	20	10	5	Mon 10:00 - 11:00
302	21	11	6	Tue 18:30 - 19:30
303	20	10	5	Thu 10:00 - 11:00

Table Description (Instructors)

A personnel directory containing names and professional specializations of the trainers employed at the facility

Table Name	Field name	Field Description	Data Type / Constraints
instructors	instructor_id	Unique identifier for each staff trainer.	Int, PK, Auto increment
	first_name	The instructor's given name.	VARCHAR(50), not null
	last_name	The instructor's family name.	VARCHAR(50), not null
	specialization	The trainer's area of expertise (e.g., "Yoga", "HIIT").	VARCHAR(30), not null

Comments on table relationships: Has a One-to-Many relationship with the **schedule** table. One instructor can be assigned to lead many different sessions in the schedule.

instructor_id	first_name	last_name	specialization
10	Sarah	Jenkins	Yoga
11	Mike	Ross	HIIT
12	Anna	Lee	Pilates

Table Description (schedule_enrollment)

A bridge table that tracks which members are registered for which scheduled classes, resolving the many-to-many relationship.

Table Name	Field name	Field Description	Data Type / Constraints
schedule_enrollment	member_id	Link to the participating member.	INT, PK, FK
	schedule_id	Link to the specific scheduled session.	INT, PK, FK

Comments on table relationships: Implements Many-to-Many between **members** and **schedule**.

member_id	schedule_id
101	301
103	301
102	302

Table Description (equipment)

An inventory of the club's physical assets, used to monitor the status and location of gym gear.

Table Name	Field name	Field Description	Data Type / Constraints

equipment	equipment_id	Unique identifier for the asset.	Int, PK, Auto increment
	name	Name of the gear (e.g., Smith Machine).	VARCHAR(100), not null
	status	Current condition (e.g., Functional).	VARCHAR(20)
	facility_id	Link to the room where it is located.	INT, FK, not null

Comments on table relationships: Many-to-One with **facilities**.

equipment_id	name	status	facility_id
701	Treadmill T80	Functional	1
702	Bench Press	Functional	1
703	Yoga Mats Kit	Functional	2

Table Description (maintenance)

A log of all repair and service activities performed on the club's equipment for resource planning.

Table Name	Field name	Field Description	Data Type / Constraints
maintenance	maintenance_id	ID for the specific repair/service event.	Int, PK, Auto increment
	maint_date	Date the maintenance was performed.	DATE, not null
	description	Technical details of the work done.	TEXT
	cost	Expense of the repair.	Decimal(10,2)

Comments on table relationships: Linked via a junction table to equipment.

maintenance_id	maintenance_date	description	cost
901	2023-12-01	Belt replacement and oiling.	150.00
902	2023-12-15	Deep cleaning and sanitizing.	40.00

Table Description (equipment_maintenance)

A junction table that records which maintenance events were performed on which specific pieces of equipment.

Table Name	Field name	Field Description	Data Type / Constraints
equipment_maintenance	equipment_id	Link to the specific piece of equipment.	INT, PK, FK
	maintenance_id	Link to the maintenance log record.	INT, PK, FK

Comments on table relationships: Resolves the Many-to-Many link between **equipment** and **maintenance** logs.

equipment_id	maintenance_id
701	901
703	902